

Kritika Pandit

Full-Stack AI Engineer

(507) 301-8587 | panditk@carleton.edu | linkedin.com/in/kritikapandit | github.com/panditk455

EXPERIENCE

Software Engineer - AgenScience AI (Seed-Stage AI Startup), Remote Apr 2026 - Present

- Designed a unified retail ontology (Merchant, Product, Order, Experiment) in Microsoft Fabric, giving pricing logic, A/B testing, and Shopify sync one consistent, governed data model instead of fragmented, drifting schemas
- Built the production Shopify backend for RevMax using Flask, SQLAlchemy, and the Admin GraphQL API, enabling merchants to run statistically valid A/B price experiments and push winning prices live in one click
- Architected a pluggable demand-model framework and benchmarked 4 pricing specifications on real transaction data, cutting elasticity-estimation failures from 10% to 0% through held-out testing

Data Engineering Intern - Oracle, Belmont, MA Nov 2025 - Apr 2026

- Architected the transition of a monolithic big data pipeline into smaller, modular products within Oracle Retail Lifecycle Pricing Optimization, boosting efficiency and reducing costs
- Secured the codebase by resolving hundreds of vulnerabilities and deploying critical patches, and designed OCI dashboards monitoring memory, latency, and KPIs using Terraform-based infrastructure as code across AWS and OCI

Software Engineer Intern - United Nations DESA, Remote Aug - Oct 2025

- Built Python ETL pipelines via GitHub Actions CI/CD, replacing a 3-day manual process with a fully automated pipeline ingesting 180+ country-level datasets into PostgreSQL for the World Investment Report 2025
- Communicated technical tradeoffs clearly to non-technical policy stakeholders across a globally distributed team, translating complex international data into decision-ready insights for real policy decisions

Student Software Developer (Part-Time) - Academic Technology, Carleton College Jun 2023 - Jun 2026

- Progressed from building ML pipelines for faculty research to owning a production Llama 3 RAG pipeline end-to-end, deployed via Flask, FAISS, and PostgreSQL to 2,000+ users
- Owned monitoring across 300+ deployed faculty WordPress sites, tracing a DMARC email deliverability failure to its root cause in DNS and SMTP configuration independently

PROJECTS

FinLoop - Self-Improving Document Analysis Agent [\[Demo\]](#) May - Jun 2026

- Built a 3-agent Claude API loop orchestrated as a LangGraph state machine, where a judge agent scores outputs and routes to re-optimization or final evaluation based on results
- Validated generalization with held-out questions (+13.0) and a held-out document from an unrelated industry (+12.1), proving the improvement wasn't domain-specific or a Goodhart's law artifact

MarketPulse - Real-Time Data Platform [\[Demo\]](#) Feb - Apr 2026

- Built an hourly self-healing reconciliation job using Airflow, independently re-checking pipeline output against raw data and catching a real production bug where server restarts silently dropped data
- Streamed live Coinbase and Kraken trade feeds through Kafka into TimescaleDB, building event-time aggregation that stayed exactly-once in effect on top of at-least-once delivery

Voice AI Acquisitions Agent [\[GitHub\]](#) Sep - Nov 2025

- Built a production agentic phone system using Twilio Voice and LLM function-calling, owning the system end-to-end from backend architecture to a client-facing React dashboard
- Classified caller intent and triggered structured downstream actions under a live client deadline, giving the client self-serve visibility into agent performance without engineering support

Secure File Transfer System [\[Demo\]](#) 2025

- Implemented a socket-level client-server protocol with RSA-2048 key exchange and AES-256-GCM encryption, writing a full threat model defining every failure state before any implementation code
- Validated the system against replay attacks and tampered packets, structuring authentication, encryption, and transfer as three independently testable modules

EDUCATION

Carleton College

B.A. Computer Science & Statistics | GPA: 3.70/4.00
Graduated with Distinction in Major: Computer Science

2022 - 2026

Northfield, MN

SKILLS

Data & Ontology: Microsoft Fabric, data modeling, ontology design, SQL, PostgreSQL, data governance, ETL pipelines
Backend & AI: Python, Flask, Claude API, LLM applications, agentic workflows, RAG pipelines, TensorFlow, PyTorch, evaluation frameworks
Frontend & Infra: TypeScript, React, Next.js, Kafka, Docker, GitHub Actions CI/CD, Azure, AWS, Git